A PSEUDO-COHERENT COMPLEX OVER A LOCAL RING IS DETERMINED UP TO ISOMORPHISM BY THE DIMENSIONS OF ITS RESIDUE FIELD FIBRES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 19547-19563 (canonical tag 0BCC) and the counted proof is the proof environment at source lines 19565-19591, which immediately follows it and is its own proof.

Editorial note. The statement is written in two sentences both beginning with Then, as printed by the author; the text is copied exactly as it stands and is not corrected.

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Allowed standard prerequisites (editorial).

- Pseudo-coherence is preserved by pulling back along the residue field map, so the fibre of the complex at the residue field is pseudo-coherent and its cohomology spaces are finite dimensional and vanish above some degree. invoked at proof lines 19565-19569. Establishes that the dimensions in the statement are finite and eventually zero, which is what the displayed complex uses.
- Every object of the derived category of a field is isomorphic to a complex with zero differentials, whose terms are the cohomology spaces. used at proof lines 19569-19572. Puts the fibre in the normal form to which the lifting statement applies.
- A map in the derived category between bounded complexes of projectives is represented by a map of complexes, and a map of finite free modules whose reduction modulo the maximal ideal is an isomorphism is an isomorphism. used at proof lines 19574-19591. Supplies the map of complexes in the uniqueness argument and upgrades its degreewise reductions to isomorphisms.

Source reference. Original lemma 0650: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16345–16345. The reference is applied at proof lines 19565-19566 to see that the fibre at the residue field is pseudo-coherent, which gives

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finiteness and vanishing of its cohomology; the rest of the existence argument is inside the counted span.

Source reference. Original lemma 0BCB: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19517–19517. The reference produces the complex of free modules representing the object from the fibre data, applied at proof lines 19572-19573; the uniqueness half of the proof is inside the counted span.

Source reference. Original lemma 064B: sources/stacks-chow/tags.txt, source lines 7860–7860. The reference lets the isomorphism between the two representatives be realized by a map of complexes, invoked at proof lines 19579-19582; the degreewise reduction argument that follows is inside the counted span.

Lemma 0.2. *Let $(R, \mathfrak{m}, \kappa)$ be a local ring. Let $K \in D(R)$ be pseudo-coherent. Set $d_i = \dim_\kappa H^i(K \otimes^{\mathbf{L}}_R \kappa)$. Then $d_i < \infty$ and for some $b \in \mathbf{Z}$ we have $d_i = 0$ for $i > b$. Then there exists a complex*

$$\ldots \to R^{\oplus d_{b-2}} \to R^{\oplus d_{b-1}} \to R^{\oplus d_b} \to 0 \to \ldots$$

representing K in $D(R)$. Moreover, this complex is unique up to isomorphism(!).

Proof. Observe that $K \otimes^{\mathbf{L}}_R \kappa$ is pseudo-coherent as an object of $D(\kappa)$, see Lemma 0650. Hence the cohomology spaces are finite dimensional and vanish above some cutoff. Every object of $D(\kappa)$ is isomorphic in $D(\kappa)$ to a complex $E^\bullet$ with zero differentials. In particular $E^i \cong \kappa^{\oplus d_i}$ is finite free. Applying Lemma 0BCB we obtain the existence.

If we have two complexes $F^\bullet$ and $G^\bullet$ with F^i and G^i free of rank d_i representing K. Then we may choose a map of complexes $\beta : F^\bullet \to G^\bullet$ representing the isomorphism $F^\bullet \cong K \cong G^\bullet$, see Derived Categories, Lemma 064B. The induced map of complexes $\beta \otimes 1 : F^\bullet \otimes^{\mathbf{L}}_R \kappa \to G^\bullet \otimes^{\mathbf{L}}_R \kappa$ must be an isomorphism of complexes as the differentials in $F^\bullet \otimes^{\mathbf{L}}_R \kappa$ and $G^\bullet \otimes^{\mathbf{L}}_R \kappa$ are zero. Thus $\beta^i : F^i \to G^i$ is a map of finite free R-modules whose reduction modulo $\mathfrak{m}$ is an isomorphism. Hence β^i is an isomorphism and we win. $\square$